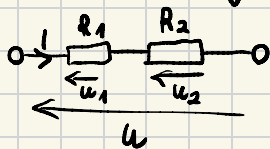


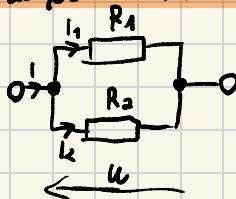
ĆWICZENIE 1

Łączenie szeregowo



$$R_z = R_1 + R_2$$

Łączenie równoległe

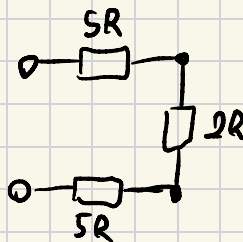
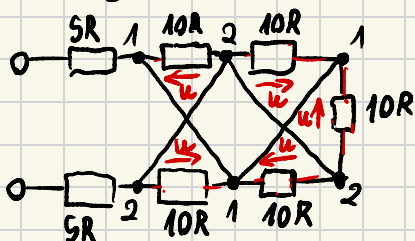


$$\frac{1}{R_z} = \frac{1}{R_1} + \frac{1}{R_2}$$

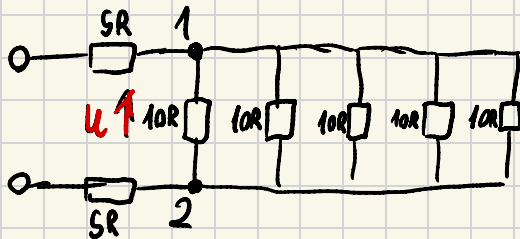
$$R_z = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Oblicz rezystancję zastępczą

b)

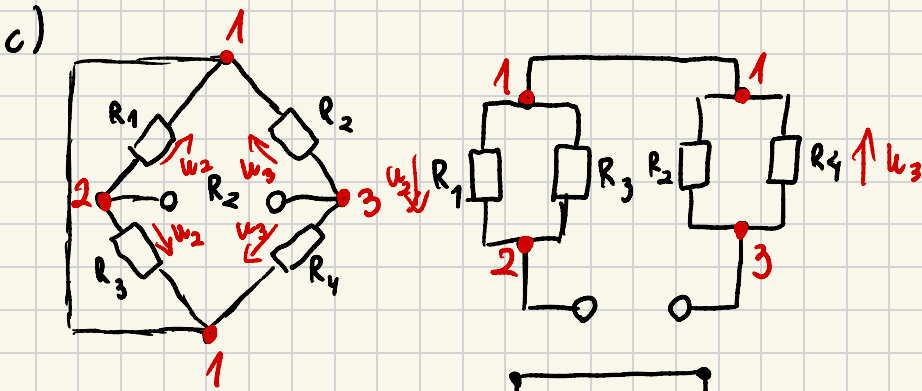


$$R_z = 5R + 5R + 2R = 12R$$



$$\frac{1}{R_{z1}} = \sum \frac{1}{10R}$$

$$\frac{1}{R_{z1}} = \frac{5}{10R} \Rightarrow R_{z1} = 2R$$



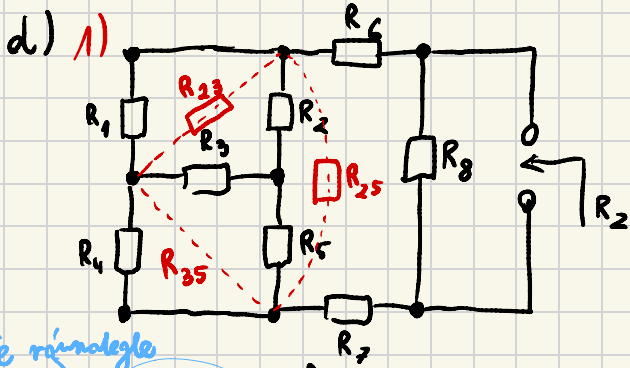
$$R_1 = R_2 = 4 \, \Omega$$

$$R_3 = R_4 = 6 \, \Omega$$

$$R_{21} = \frac{R_1 \cdot R_3}{R_1 + R_3} = \frac{4 \cdot 6}{4 + 6} = \frac{24}{10} = 2,4 \, \Omega$$

$$R_{22} = \frac{R_2 \cdot R_4}{R_2 + R_4} = \frac{4 \cdot 6}{4 + 6} = \frac{24}{10} = 2,4 \, \Omega$$

$$R_2 = R_{21} + R_{22} = 4,8 \, \Omega$$



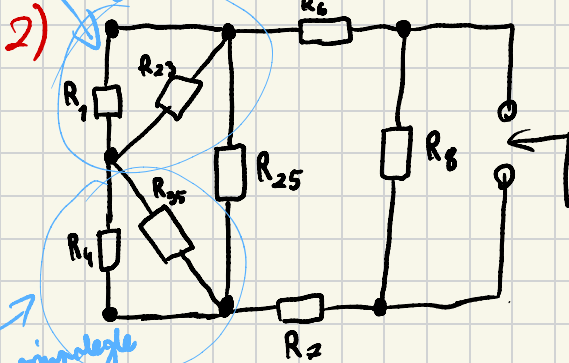
$$R_1 = R_4 = 9 \Omega$$

$$R_2 = R_3 = R_5 = 3 \Omega$$

$$R_6 = 0,5 \Omega \quad R_7 = 1 \Omega$$

$$R_8 = 4 \Omega$$

Te r  nolegle

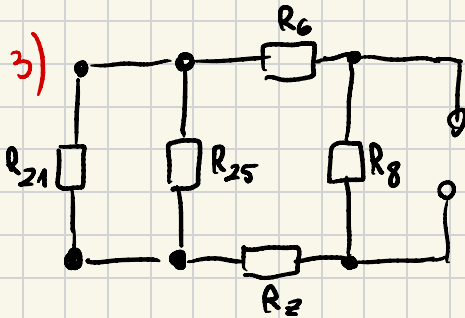


$$R_{23} = R_2 + R_3 + \frac{R_2 \cdot R_3}{R_5} = 6 + \frac{9}{3} = 9 \Omega$$

$$R_{35} = R_{25} = R_{23} = 9 \Omega$$

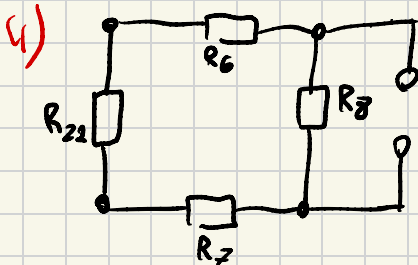
$$R_{21} = \frac{R_1 \cdot R_{23}}{R_1 + R_{23}} + \frac{R_4 \cdot R_{35}}{R_4 + R_{35}} = \frac{9 \cdot 9}{9 + 9} + \frac{9 \cdot 9}{9 + 9} = \frac{81}{18} + \frac{81}{18} = \frac{9}{2} + \frac{9}{2} = 9 \Omega$$

$$R_{22} = \frac{R_{21} \cdot R_{25}}{R_{21} + R_{25}} = \frac{9 \cdot 9}{9 + 9} = 4,5 \Omega$$



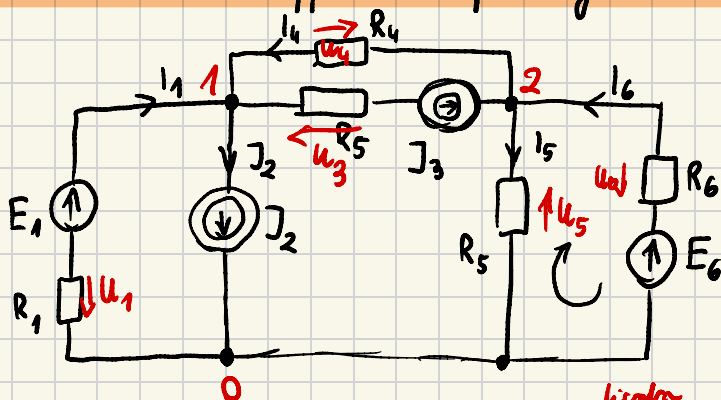
$$R_z = \frac{(R_6 + R_{22} + R_7) \cdot R_8}{(R_6 + R_{22} + R_7) + R_8} =$$

$$= \frac{6 \cdot 4}{6 + 4} = \frac{24}{10} = 2,4 \Omega$$



Napisac równania Kirchhoffa dla podanego obwodu

a)



Szukane: I_1, I_4, I_5, I_6

$n = 3$ ← liczn węzłów z różnymi potencjałami

$b = 4$ ← liczn gałęzi

$PPK = n - 1 = 2$

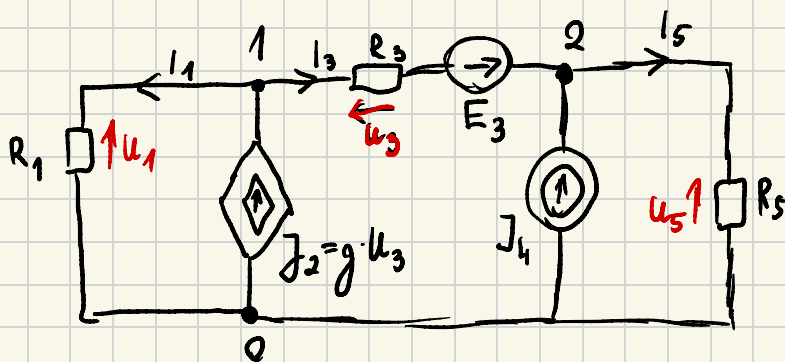
$NPK = b - n + 1 = 2$

1) $I_1 - J_2 - J_3 + I_4 = 0$

1) $R_5 I_5 + R_6 I_6 - E_6 = 0$

2) $J_3 - I_4 + I_6 - I_5 = 0$

2) $-R_1 I_1 + E_1 + R_4 I_4 - R_5 I_5 = 0$

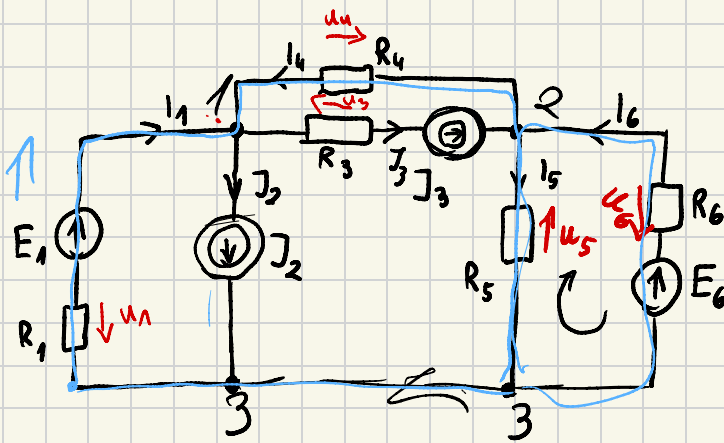


$n = 3 \quad b = 3$

$J_2 - I_1 - I_3 = 0$

$PPK (n - 1 = 2)$

$NPK (b - n + 1 = 1)$



$$n = 3 \quad b = 4$$

$$\text{PPK}(n-1=2)$$

$$\text{NPK}(b-n+1=2)$$

$$E_1 + u_4 - u_5 - u_1 = 0$$

$$E_6 - u_6 - u_5 = 0$$

$$1) \begin{cases} I_1 + I_4 = J_2 + J_3 \\ I_1 + I_4 - J_2 - J_3 = 0 \end{cases}$$

$$J_3 + I_6 = I_4 + I_5$$

$$2) J_3 + I_6 - I_4 - I_5 = 0$$